

Sta 160 Project

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```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    3.5.1      v tibble     3.2.1
## v lubridate  1.9.3      v tidyr      1.3.1
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(palmerpenguins)
```

```
data("penguins")
```

```
head(penguins)
```

```
## # A tibble: 6 x 8
##   species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>         <dbl>         <dbl>             <int>         <int>
## 1 Adelie Torgersen         39.1           18.7             181          3750
## 2 Adelie Torgersen         39.5           17.4             186          3800
## 3 Adelie Torgersen         40.3            18             195          3250
## 4 Adelie Torgersen          NA            NA              NA           NA
## 5 Adelie Torgersen         36.7           19.3             193          3450
## 6 Adelie Torgersen         39.3           20.6             190          3650
## # i 2 more variables: sex <fct>, year <int>
```

```
nrow(penguins)
```

```
## [1] 344
```

```
colSums(is.na(penguins))
```

```
##           species           island bill_length_mm bill_depth_mm
##           0              0           2              2
## flipper_length_mm body_mass_g      sex           year
##           2              2          11              0
```

```
penguins_clean <- penguins %>%
```

```
drop_na()
```

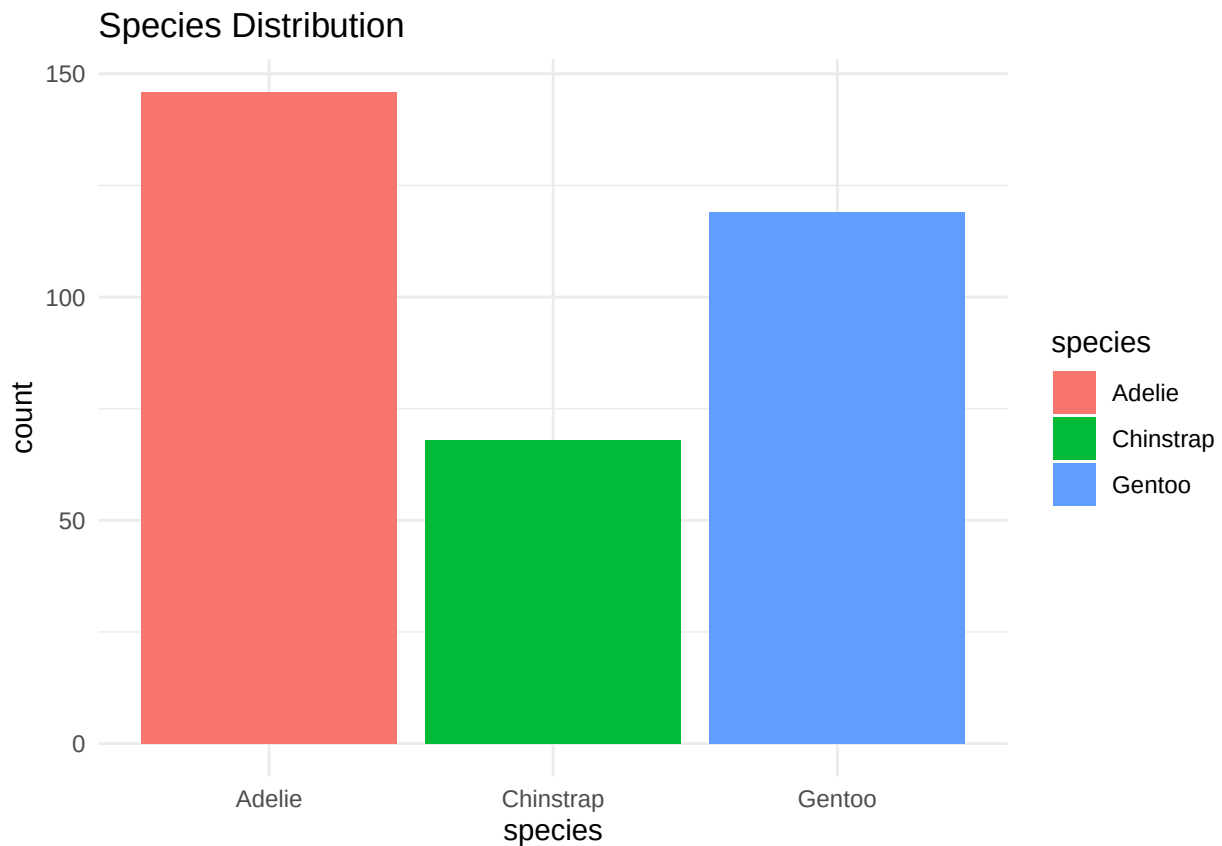
```
nrow(penguins_clean)
```

```
## [1] 333
```

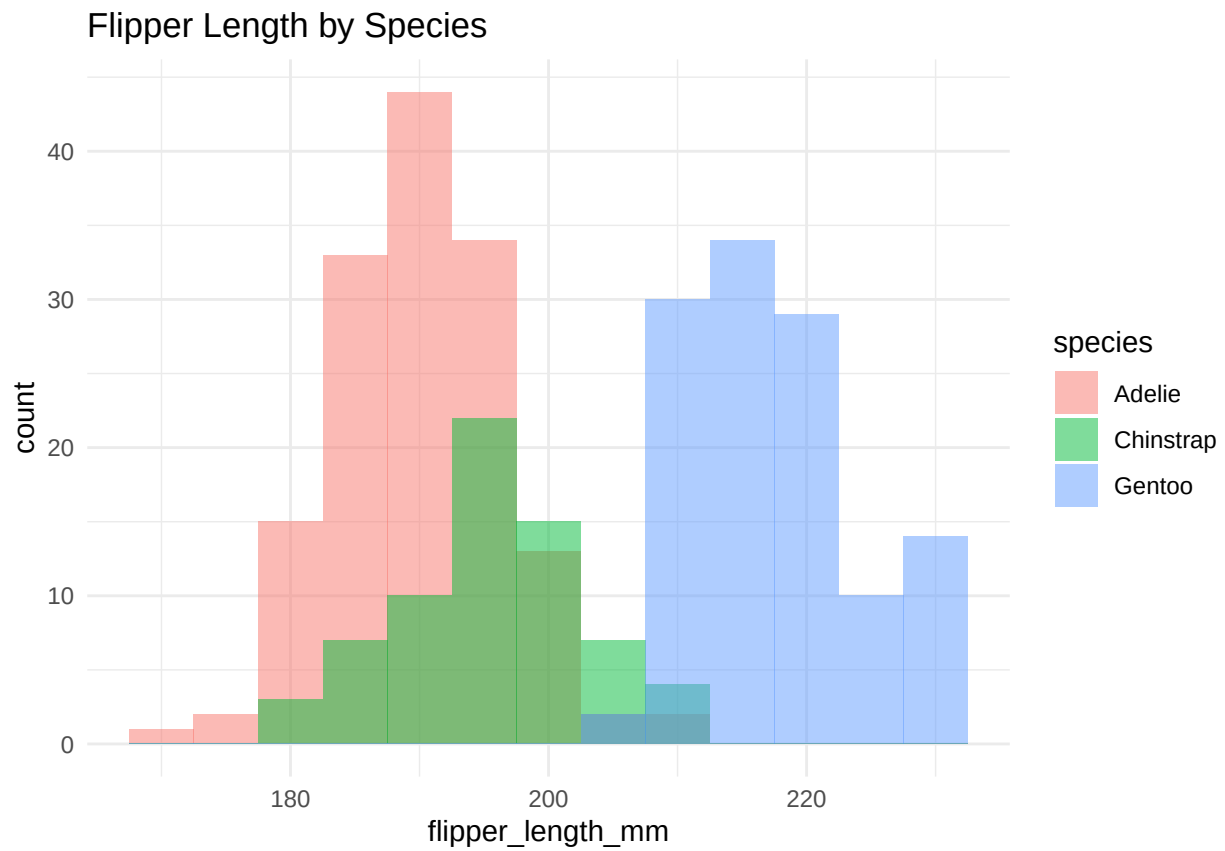
```
summary(penguins_clean)
```

```
##      species      island bill_length_mm bill_depth_mm
## Adelie   :146  Biscoe   :163   Min.    :32.10   Min.    :13.10
## Chinstrap: 68  Dream    :123   1st Qu.:39.50   1st Qu.:15.60
## Gentoo   :119  Torgersen: 47   Median :44.50   Median :17.30
##                                     Mean    :43.99   Mean    :17.16
##                                     3rd Qu.:48.60   3rd Qu.:18.70
##                                     Max.    :59.60   Max.    :21.50
## flipper_length_mm body_mass_g      sex      year
## Min.    :172      Min.    :2700  female:165  Min.    :2007
## 1st Qu.:190      1st Qu.:3550  male  :168  1st Qu.:2007
## Median :197      Median :4050                Median :2008
## Mean    :201      Mean    :4207                Mean    :2008
## 3rd Qu.:213      3rd Qu.:4775                3rd Qu.:2009
## Max.    :231      Max.    :6300                Max.    :2009
```

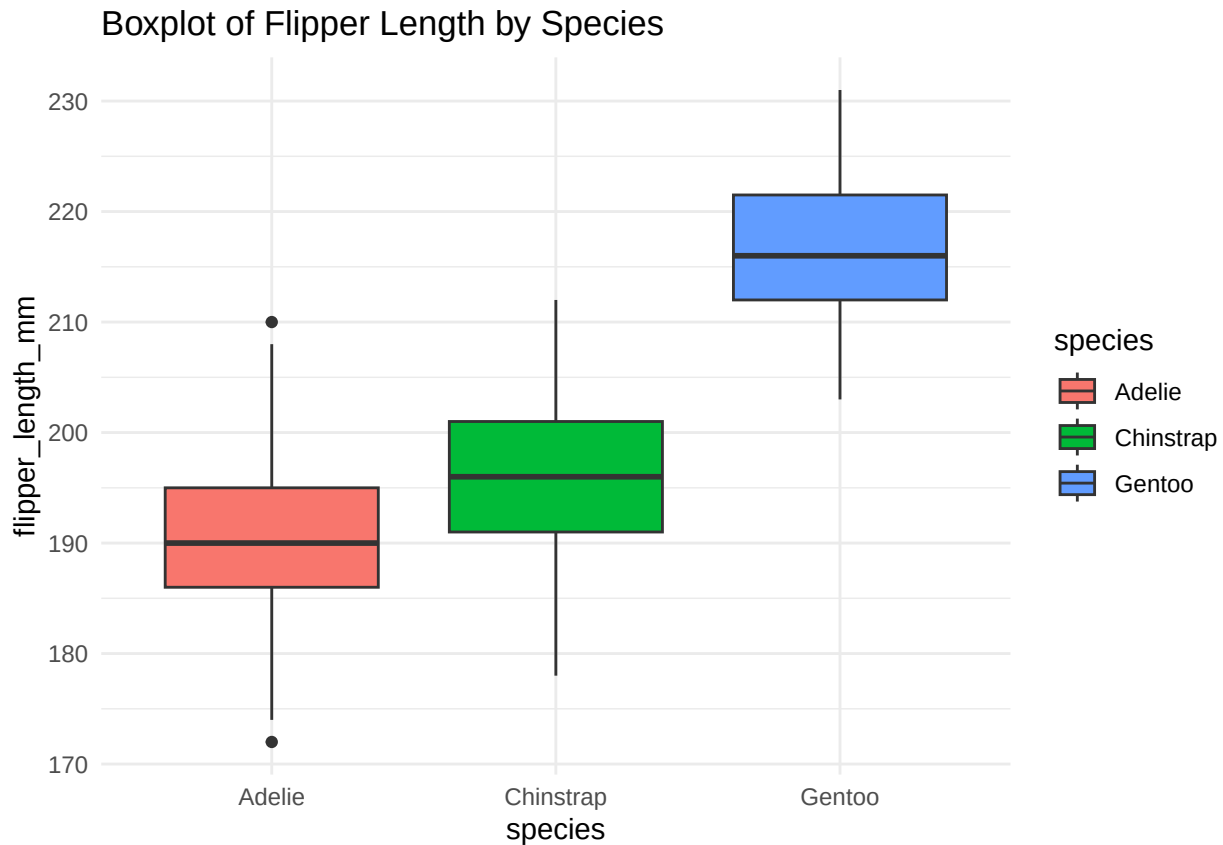
```
ggplot(penguins_clean, aes(x = species, fill = species)) +
  geom_bar() +
  theme_minimal() +
  labs(title = "Species Distribution")
```



```
ggplot(penguins_clean, aes(x = flipper_length_mm, fill = species)) +
  geom_histogram(alpha = 0.5, position = "identity", binwidth = 5) +
  theme_minimal() +
  labs(title = "Flipper Length by Species")
```



```
ggplot(penguins_clean, aes(x = species, y = flipper_length_mm, fill = species)) +  
  geom_boxplot() +  
  theme_minimal() +  
  labs(title = "Boxplot of Flipper Length by Species")
```



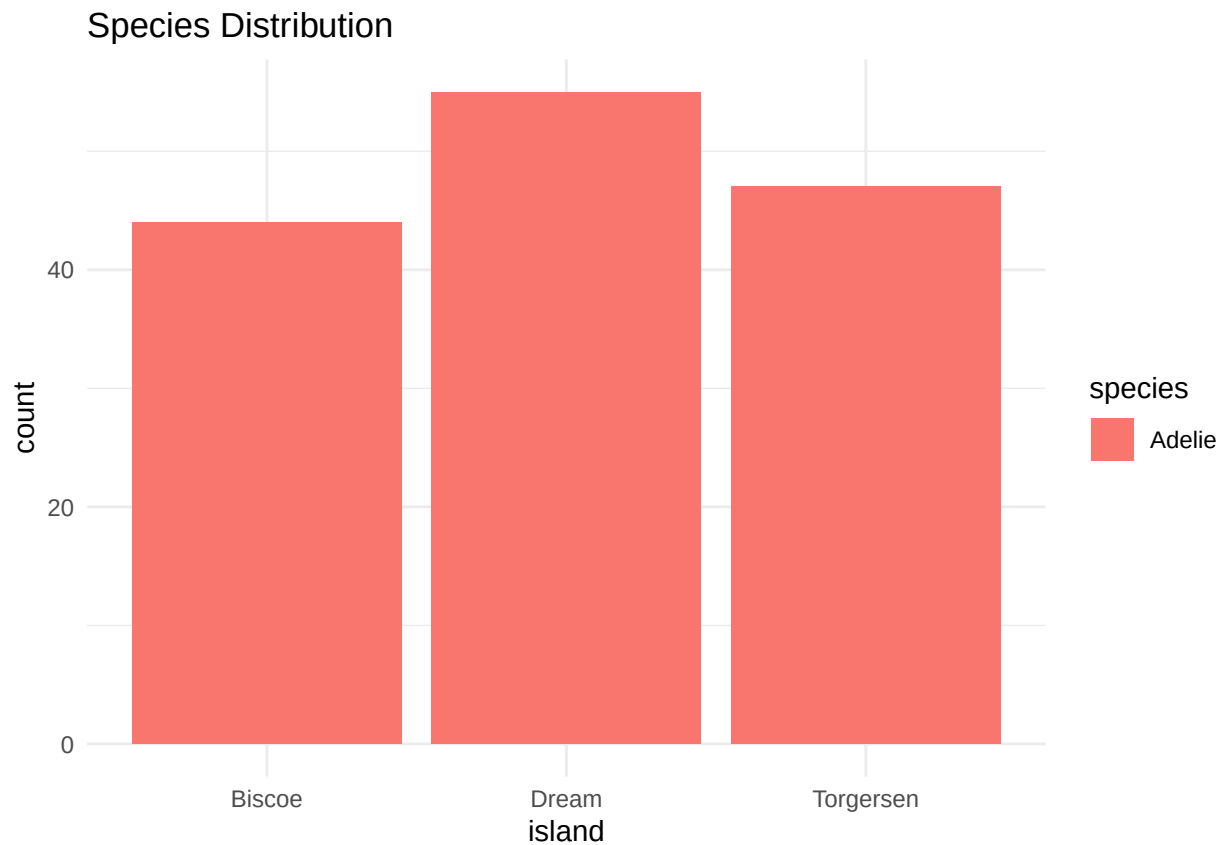
How are Adelli Penguins Different across isalnds? How do their size change? Is Gender a factor?

```
adelie_tbl <- penguins_clean |>
  filter(species == "Adelie")
```

```
adelie_tbl
```

```
## # A tibble: 146 x 8
##   species island   bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>         <dbl>         <dbl>         <int>         <int>
## 1 Adelie  Torgersen         39.1          18.7          181          3750
## 2 Adelie  Torgersen         39.5          17.4          186          3800
## 3 Adelie  Torgersen         40.3           18          195          3250
## 4 Adelie  Torgersen         36.7          19.3          193          3450
## 5 Adelie  Torgersen         39.3          20.6          190          3650
## 6 Adelie  Torgersen         38.9          17.8          181          3625
## 7 Adelie  Torgersen         39.2          19.6          195          4675
## 8 Adelie  Torgersen         41.1          17.6          182          3200
## 9 Adelie  Torgersen         38.6          21.2          191          3800
## 10 Adelie Torgersen         34.6          21.1          198          4400
## # i 136 more rows
## # i 2 more variables: sex <fct>, year <int>
```

```
ggplot(adelie_tbl, aes(x = island, fill = species)) +
  geom_bar() +
  theme_minimal() +
  labs(title = "Species Distribution")
```



```

adelie_summary <- adelic_tbl |>
  group_by(island) |>
  summarise(
    mean_bill_length_mm = mean(bill_length_mm, na.rm = TRUE),
    mean_bill_depth_mm = mean(bill_depth_mm, na.rm = TRUE),
    mean_flipper_length = mean(flipper_length_mm, na.rm = TRUE),
    mean_body_mass_g = mean(body_mass_g, na.rm = TRUE),
    sd_flipper = sd(flipper_length_mm),
    sd_bill_length = sd(bill_length_mm),
    sd_bill_depth = sd(bill_depth_mm),
    sd_body_mass_g = sd(body_mass_g)
  )

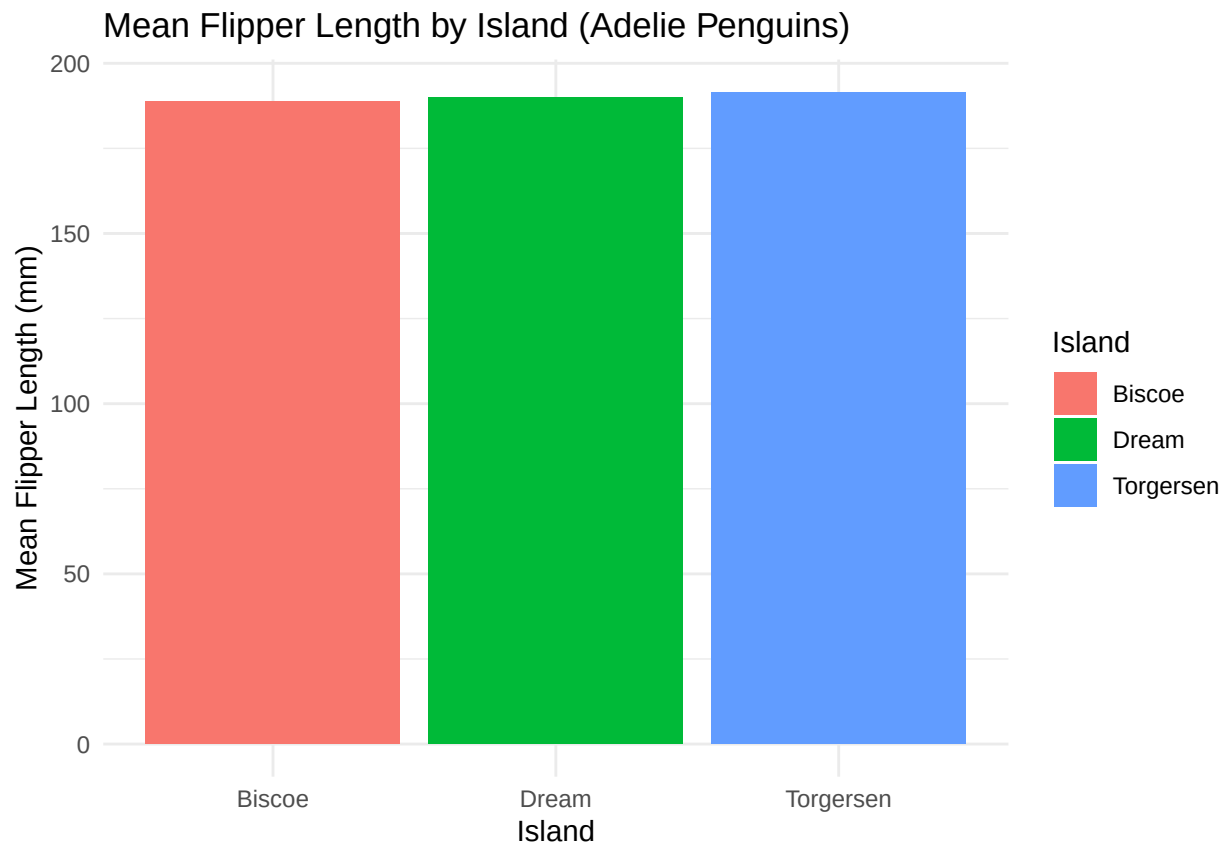
head(adelie_summary)

## # A tibble: 3 x 9
##   island      mean_bill_length_mm mean_bill_depth_mm mean_flipper_length
##   <fct>          <dbl>          <dbl>          <dbl>
## 1 Biscoe          39.0          18.4          189.
## 2 Dream           38.5          18.2          190.
## 3 Torgersen       39.0          18.5          192.
## # i 5 more variables: mean_body_mass_g <dbl>, sd_flipper <dbl>,
## #   sd_bill_length <dbl>, sd_bill_depth <dbl>, sd_body_mass_g <dbl>

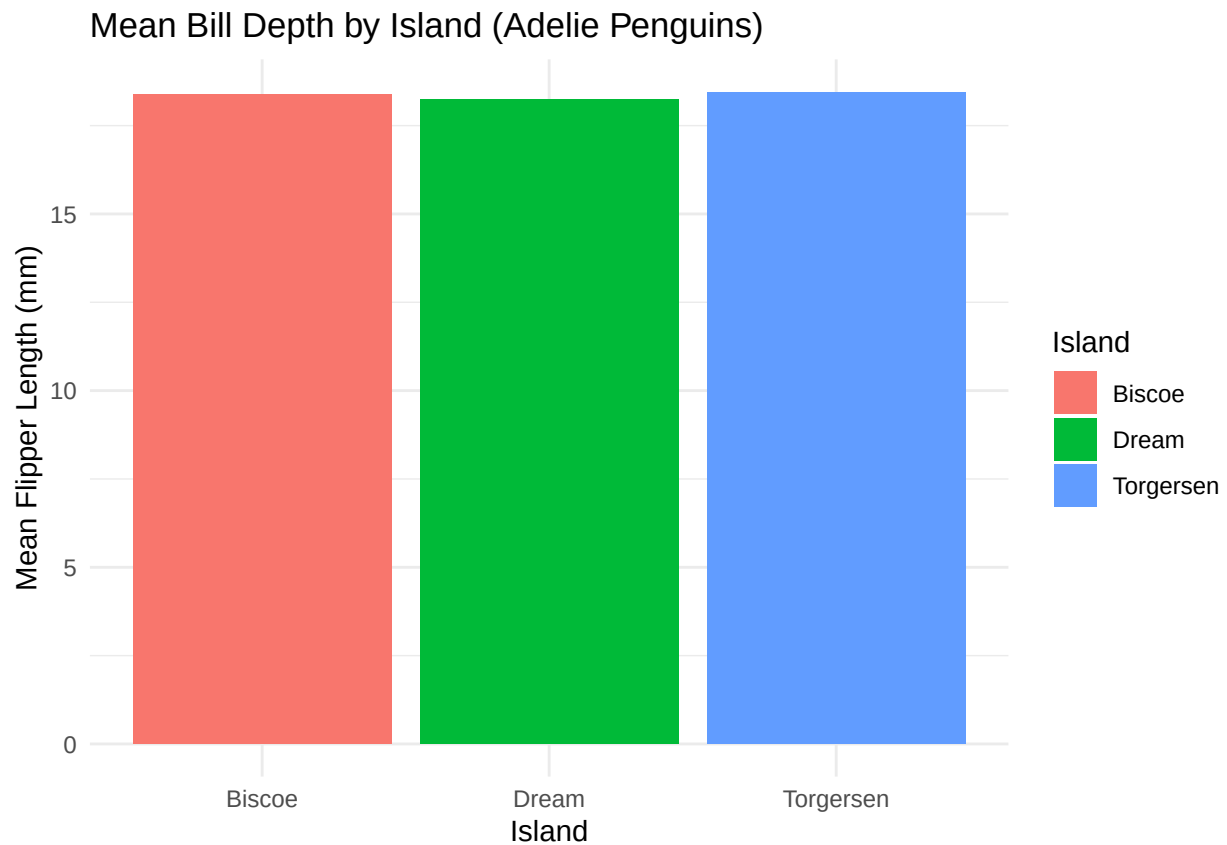
ggplot(adelie_summary, aes(x = island, y = mean_flipper_length, fill = island)) +
  geom_col() +

```

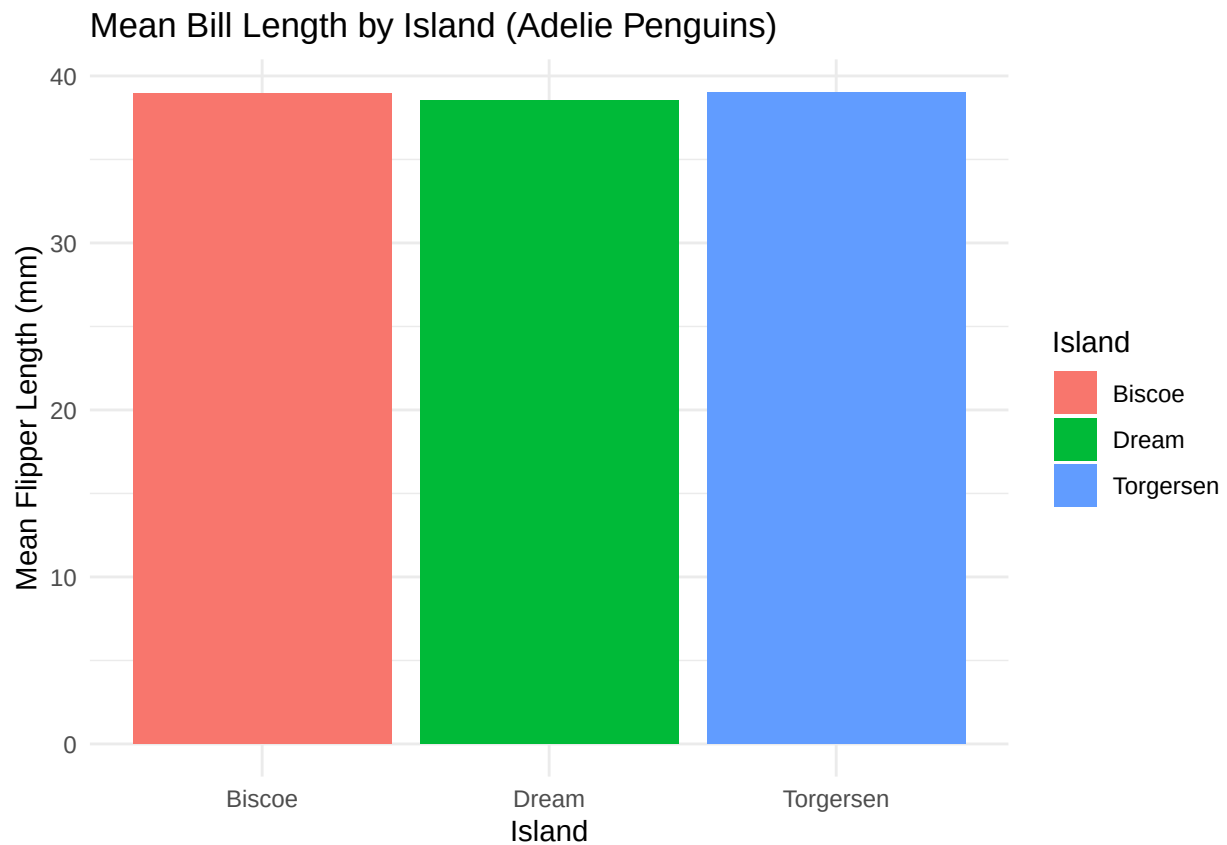
```
theme_minimal() +
labs(
  title = "Mean Flipper Length by Island (Adelie Penguins)",
  x = "Island",
  y = "Mean Flipper Length (mm)",
  fill = "Island"
)
```



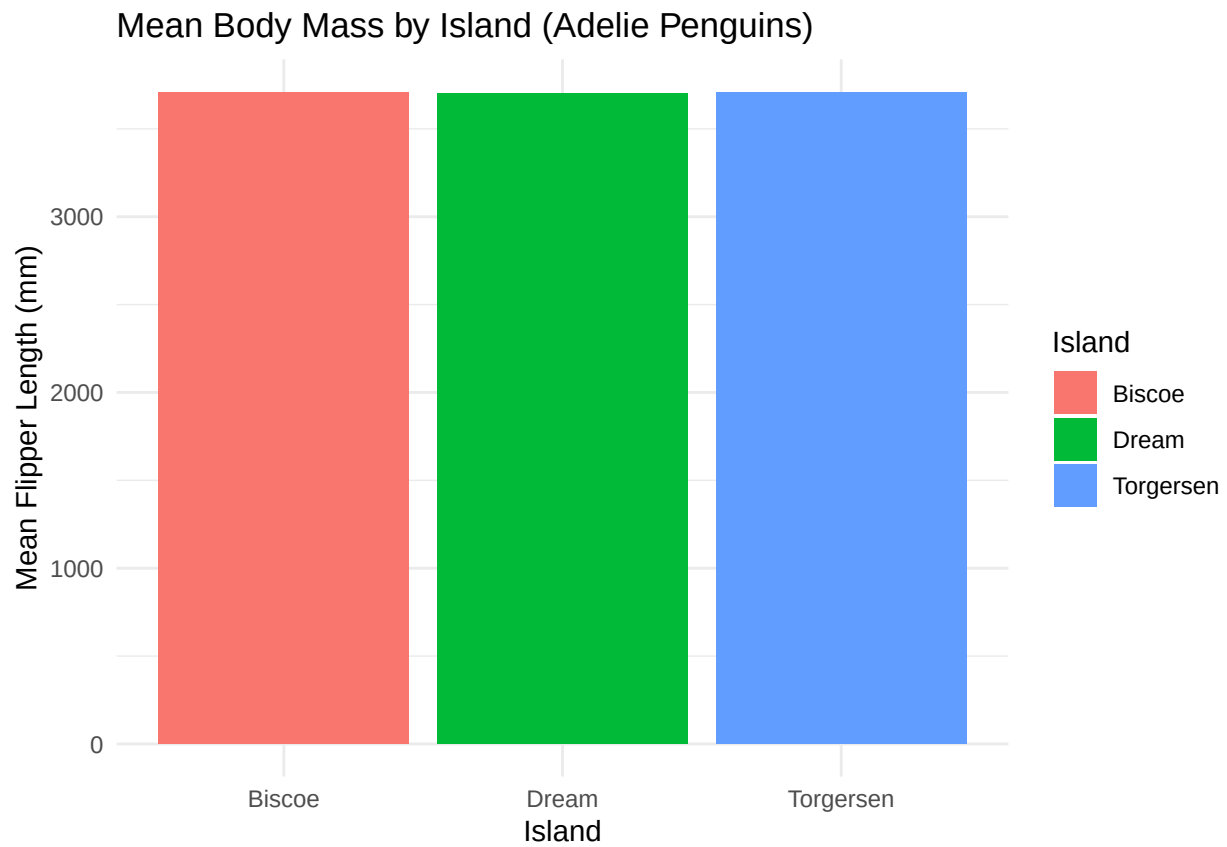
```
ggplot(adelie_summary, aes(x = island, y = mean_bill_depth_mm, fill = island)) +
  geom_col() +
  theme_minimal() +
  labs(
    title = "Mean Bill Depth by Island (Adelie Penguins)",
    x = "Island",
    y = "Mean Flipper Length (mm)",
    fill = "Island"
  )
```



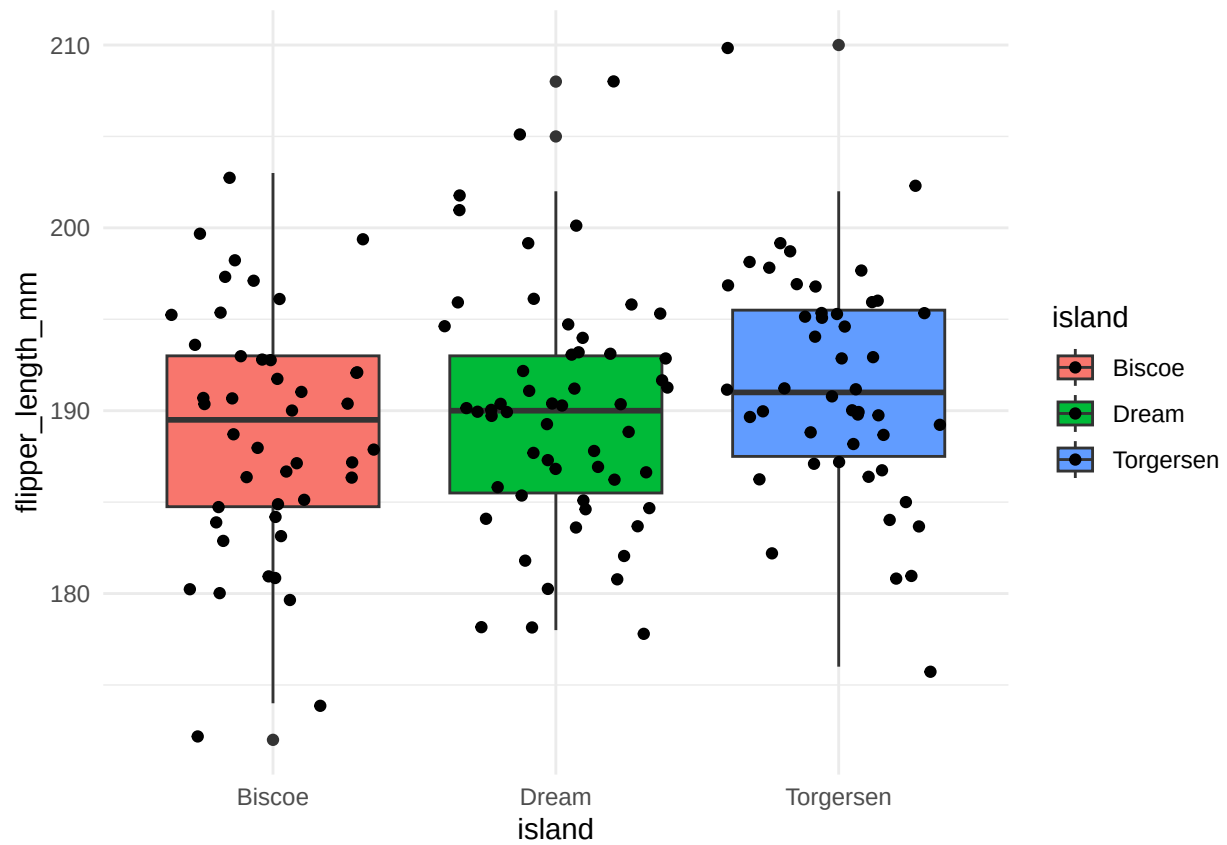
```
ggplot(adelie_summary, aes(x = island, y = mean_bill_length_mm, fill = island)) +  
  geom_col() +  
  theme_minimal() +  
  labs(  
    title = "Mean Bill Length by Island (Adelie Penguins)",  
    x = "Island",  
    y = "Mean Flipper Length (mm)",  
    fill = "Island"  
  )
```



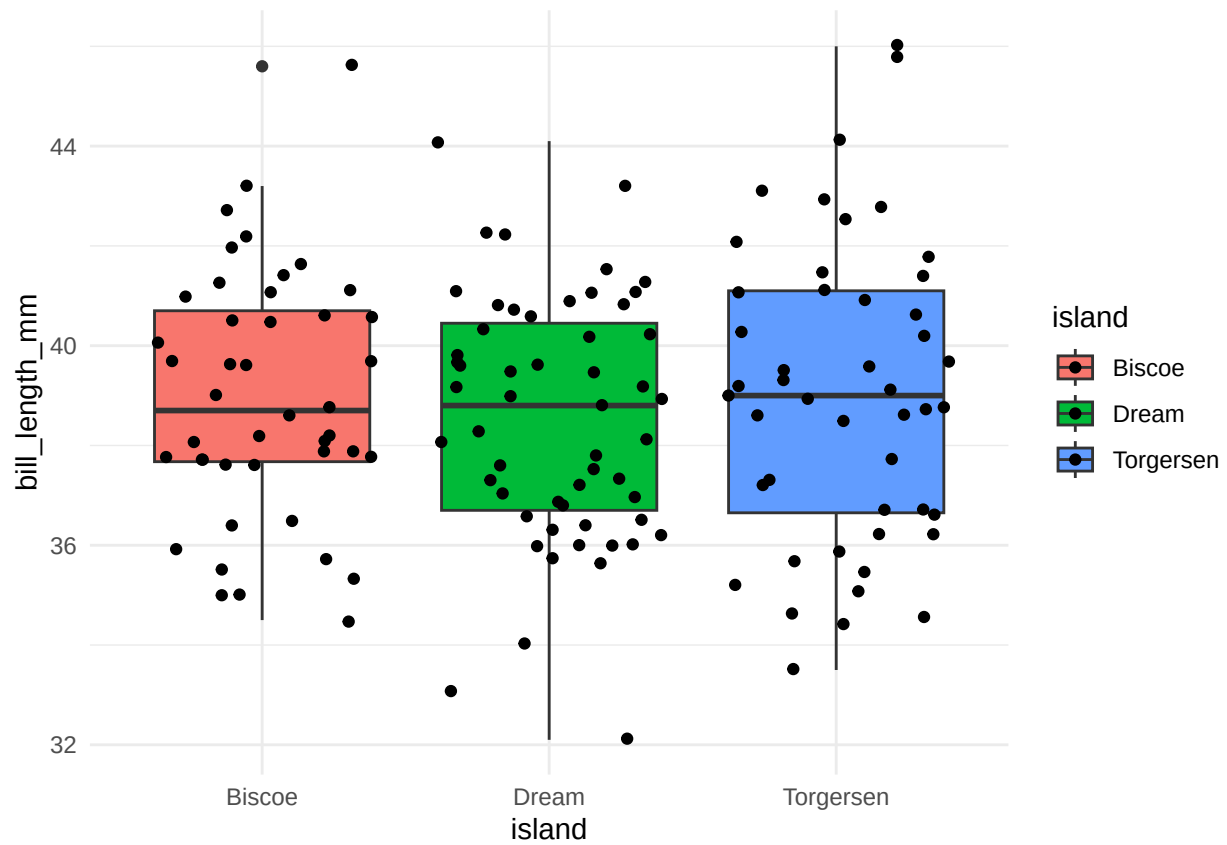
```
ggplot(adelie_summary, aes(x = island, y = mean_body_mass_g, fill = island)) +  
  geom_col() +  
  theme_minimal() +  
  labs(  
    title = "Mean Body Mass by Island (Adelie Penguins)",  
    x = "Island",  
    y = "Mean Flipper Length (mm)",  
    fill = "Island"  
  )
```



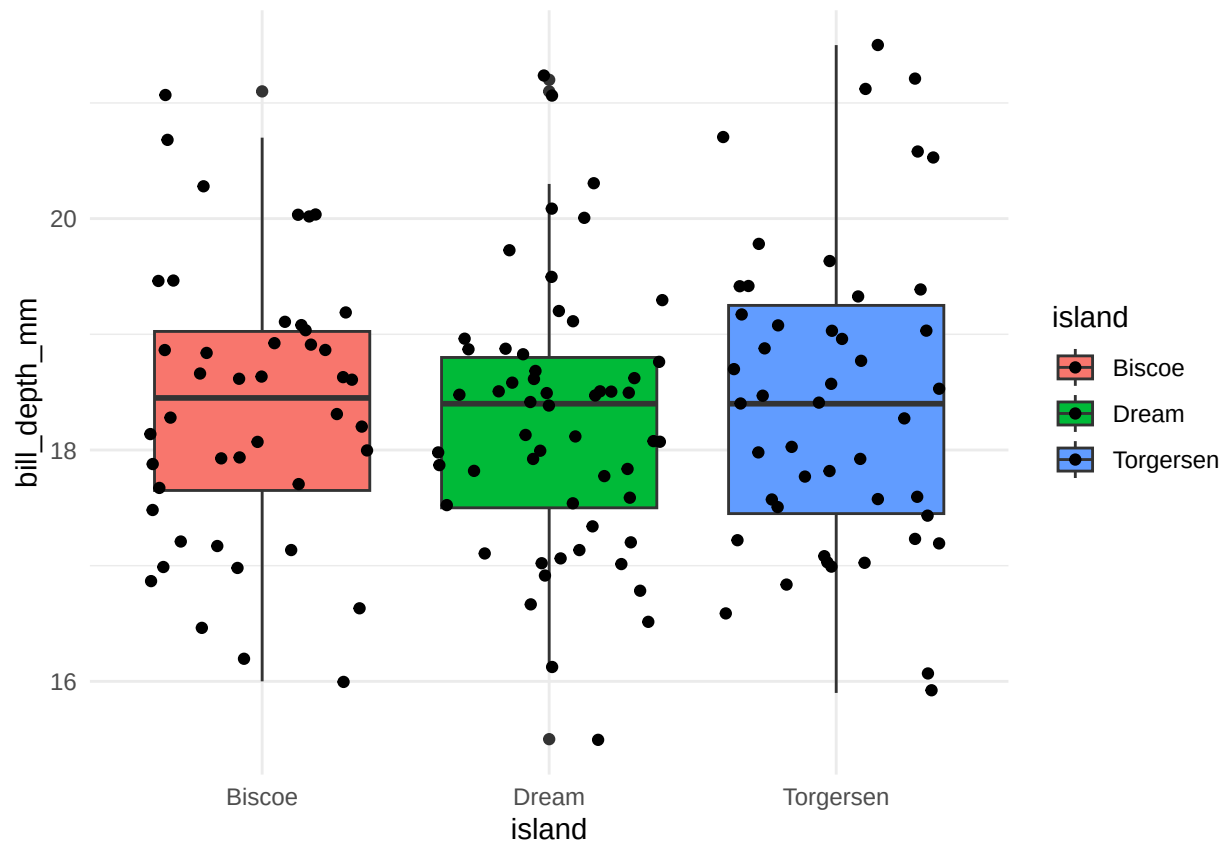
```
ggplot(adelie_tbl, aes(x = island, y = flipper_length_mm, fill = island)) +  
  geom_boxplot() +  
  geom_jitter() + # shows individual data points  
  theme_minimal()
```



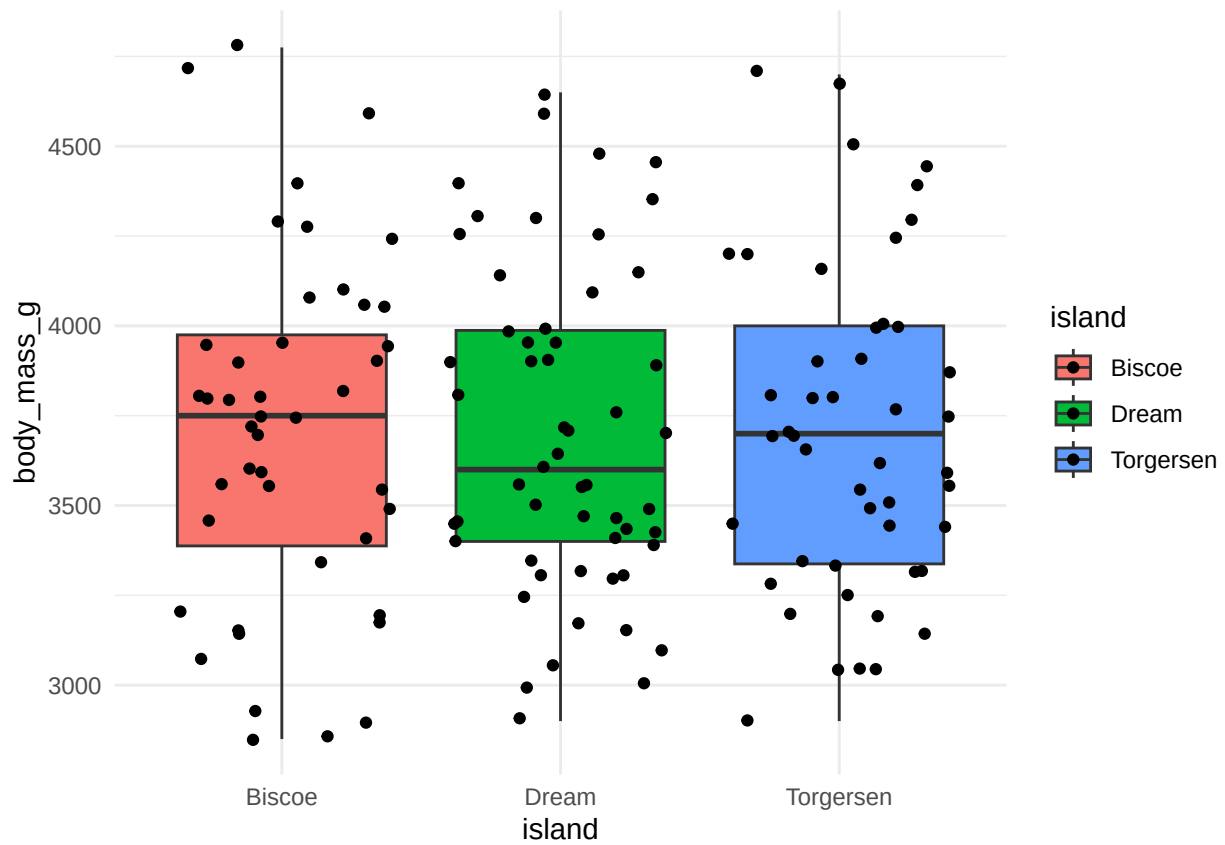
```
ggplot(adelie_tbl, aes(x = island, y = bill_length_mm, fill = island)) +
  geom_boxplot() +
  geom_jitter() + # shows individual data points
  theme_minimal()
```



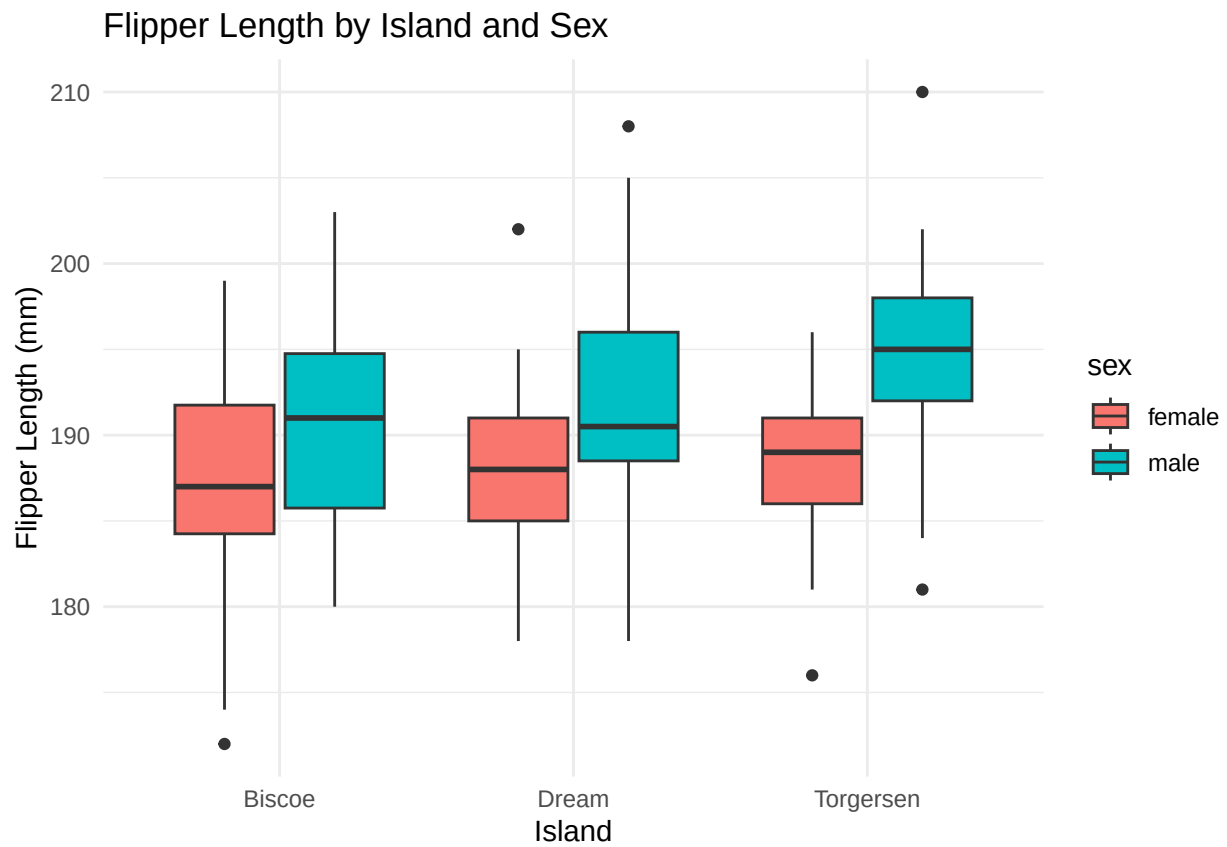
```
ggplot(adelie_tbl, aes(x = island, y = bill_depth_mm, fill = island)) +  
  geom_boxplot() +  
  geom_jitter() + # shows individual data points  
  theme_minimal()
```



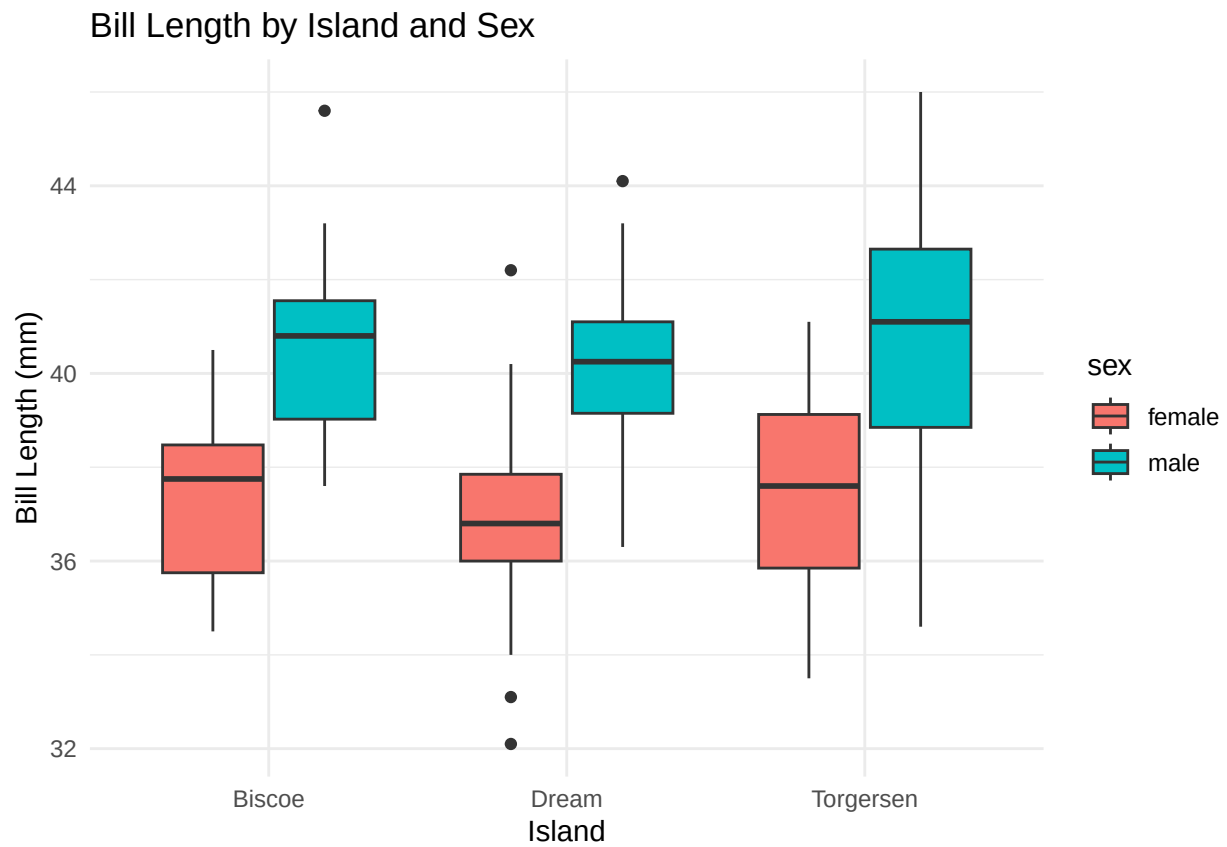
```
ggplot(adelle_tbl, aes(x = island, y = body_mass_g, fill = island)) +
  geom_boxplot() +
  geom_jitter() + # shows individual data points
  theme_minimal()
```



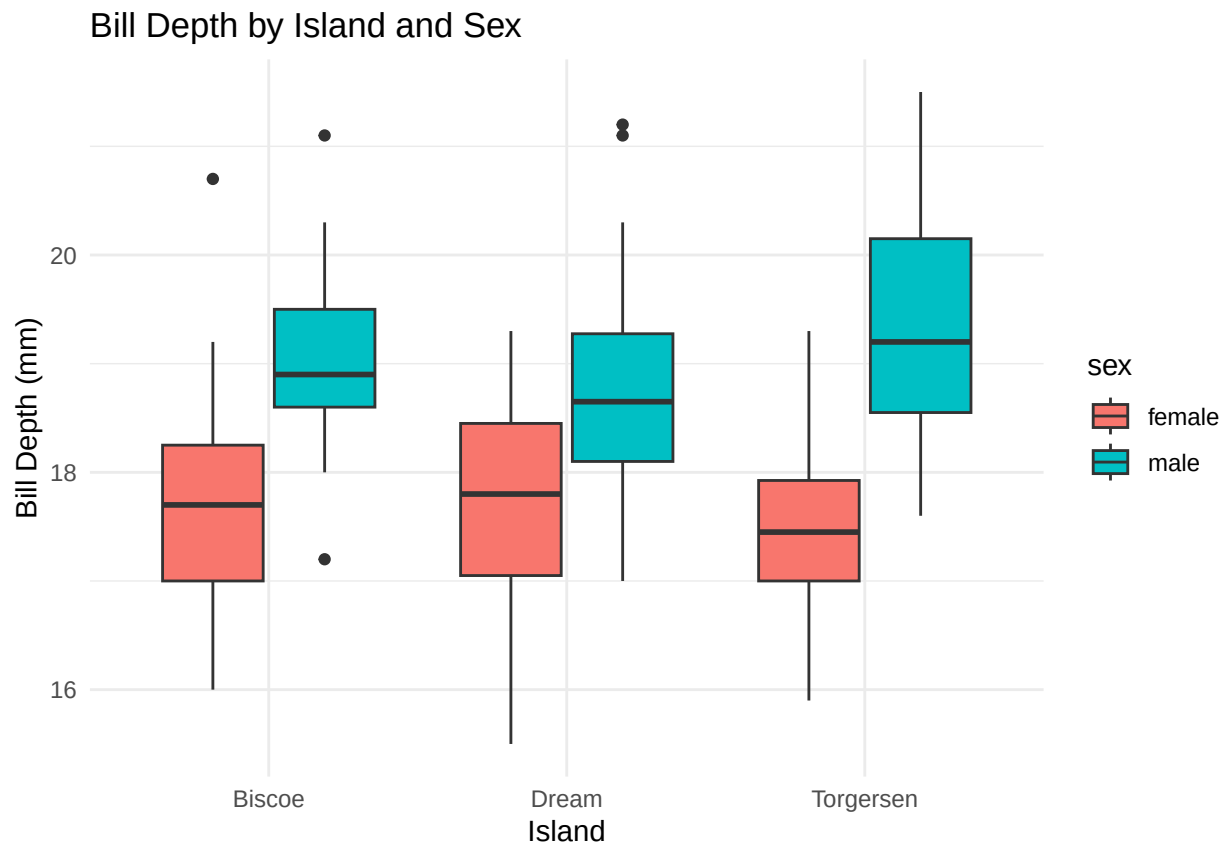
```
ggplot(adelie_tbl, aes(x = island, y = flipper_length_mm, fill = sex)) +
  geom_boxplot() +
  theme_minimal() +
  labs(title = "Flipper Length by Island and Sex",
        x = "Island", y = "Flipper Length (mm)")
```



```
ggplot(adelle_tbl, aes(x = island, y = bill_length_mm, fill = sex)) +  
  geom_boxplot() +  
  theme_minimal() +  
  labs(title = "Bill Length by Island and Sex",  
        x = "Island", y = "Bill Length (mm)")
```



```
ggplot(adelie_tbl, aes(x = island, y = bill_depth_mm, fill = sex)) +  
  geom_boxplot() +  
  theme_minimal() +  
  labs(title = "Bill Depth by Island and Sex",  
        x = "Island", y = "Bill Depth (mm)")
```



```
ggplot(adelie_tbl, aes(x = island, y = body_mass_g, fill = sex)) +  
  geom_boxplot() +  
  theme_minimal() +  
  labs(title = "Body Mass by Island and Sex",  
        x = "Island", y = "Body Mass (g)")
```

